Non-technical user feedback

Questionnaire

Question	Answer 1	Answer 2	Answer 3
What is your name?	Charlie Bull	Danny Shermer	Drew McCracken
What is your age?	Under 18	Under 18	Under 18
Were you able to navigate around the prototype?	yes	Yes	Yes
Did the prototype allow you to create an account?	Yes	Yes	Yes
How clear were the password requirements on the Register form? **(1-5)**	3	5	2
Did the prototype allow you to sign in?	Yes	Yes	Yes
Were you able to access the account dashboard?	Yes	Yes	Yes
Was the information on the account dashboard displayed clearly?	No	Yes	Yes
Were you able to book a hotel stay?	Yes	Yes	Yes
How easy was it to book a hotel stay? **(1-10)**	5	9	8
(Hotel stay) Did you get a confirmation email?	Yes	No	No
Did your hotel booking(s) appear on the account dashboard?	Yes	Yes	Yes
Were you able to book zoo tickets?	|Yes	Yes	Yes
How easy was it to book zoo tickets? **(1-10)**	5	9	5
(Zoo tickets) Did you get a confirmation email?	Yes	No	No
Did your zoo ticket(s) appear on the account dashboard?	Yes	Yes	Yes
Describe your overall booking experience	The dashboard broke after booking the zoo tickets (link to screenshot attached), also it let me book zoo tickets for 2001 which is 25 years ago and I don't have a time machine.	It was very easy and straightforward to book tickets, however there was no information in terms of costs as far as I am aware (I'm assuming the information just hasn't been added yet) .	The education page has a 404 error. the font is poor and doesn't make things clear. when trying to go to one of the attractions it just takes you to the top of the page. there shouldn't be

	https://nulc-my.sharepoint.com/:i:/g/personal/c247318_stafford_ac_uk/IQBGGw_Z2Qt9R7c_B3V367XHAb946GTaEiHriFO7_v-hcls?e=UF55Xe	Apart from that, a very smooth, very fast and very positive booking experience.	attractions on the first page if you have attractions on its own page
Did you use the prototype on a computer (desktop, laptop)?	Yes	Yes	Yes
Overall, how was your experience using the prototype on computer?	The font wasn't very readable, please improve.	It was very good. The website loaded correctly and all functioned correctly as far as I can tell.	the layout seems jarring. its felt a little bit confusing
Did you use the prototype on a mobile device (phone, tablet)?	No	No	No
Overall, how was your experience using the prototype on a mobile device?			
Were there any accessibility settings you could access?	No	No	No
Describe your experience with the accessibility settings available to you			
In your opinion, how appropriate was the colour theme of the prototype? **(1-5)**	3	3	3
How intuitive are the layout and controls of the prototype? **(1-5)**	1	4	2
On a scale of 1 to 10, how would you rate the responsiveness of the prototype? **(1-10)**	7	9	7
Overall, how would you rate your experience using the Riget Zoo Adventures website prototype? **(1-5)**	2	4	2

Conclusion

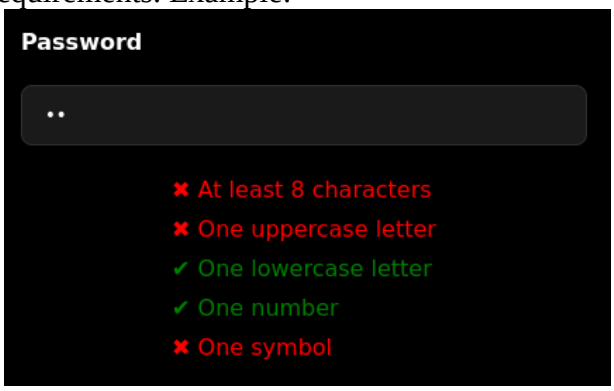
Observation of user feedback

After evaluating non-technical user feedback, the overall consensus was generally leaning positive, with a 3.8/5 star average on colour appropriateness, 8.0/10 for responsiveness, and 3.3/5 average for overall experience.

However, over 30% of respondents reported issues relating to the font style making text content difficult to read; and 30% also reported issues relating to accounts or the account dashboard, where they either could not create an account due to client-side security measures, logging out of their account causing error messages, and booking zoo tickets occasionally breaking the account login/sign-up page. One user also reported being able to book zoo tickets for 25 years in the past, this validation mistake was able to be easily reproduced, during testing the issue also applied to the hotel booking form if the client were to disable JavaScript, as validation was not applied on the server side. Another user mentioned being able to book a hotel for thousands of years in the future (02/02/9999), this is technically not invalid, however their ability to attend is questionable.

Changes required

Issue	Solution
Font style was generally difficult to read	Update the default font style to one that is easier to read, and renders the same regardless of setup
Users did not receive confirmation emails for hotel bookings or purchasing tickets	Update the system so that users are sent a confirmation email with their booking details whenever they book a hotel or purchase zoo tickets
Users able to book ticket and hotel stays in the past	Update the ticket booking form to have client-side (front-end) validation, and update both the ticket and hotel booking forms to include server-side (back-end) validation
Users able to book for thousands of years in the future	While technically not an invalid booking date, this should be fixed by adding client-side and server-side validation to only allow bookings up to 2 years in advance
Hotel and ticket booking pages do not display a price to the user	Update the booking forms to include calculations for how much their hotel/ticket booking will cost them, as well as methods to pay for their booking
Educational visits and resources page has no content/returns a 404 (not found) response	Update the solution to include resources for educational visits, as this has not yet been implemented in the current solution
Some users unable to make accounts with certain antivirus software	Update the solution to include a valid TLS certificate from a trusted authority (such as Cloudflare) so the solution can use HTTPS (encrypted web traffic), this should remove any security warnings
Dashboard occasionally breaks after booking tickets, claiming headers have already been sent	Update the process that redirects users to the account dashboard after booking to include thorough checks to ensure headers are not sent multiple times, and users arrive as intended

Clicking on the cards on the attractions page takes the user to the top of the same page, instead of more information about the attraction	Create separate pages for each attraction that holds more information, and update URLs on the attraction cards to link to their corresponding pages
The website layout breaks, clips over itself, and extends past the width of the screen on mobile devices	Update the solution to include dedicated formatting for mobile layouts, to ensure compatibility and accessibility
Users were unable to utilise any accessibility settings	Update the solution to include an accessibility menu on each page. This menu should include options for increasing font size, changing font colour/style, increased contrast, reduced motion, ensuring all media elements have accurate Alt-text
Password requirements on the sign-up form are not made clear to the user before they start typing	Update the sign-up form to show the user the password requirements, as well as a live update showing them their met/unmet password requirements. Example: 

Technical user feedback

Comprehensive/peer code review

Black-box testing

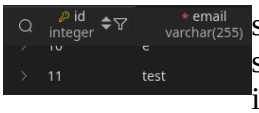
Technical users were asked to verify:

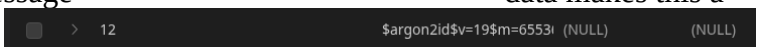
- User interface – ensuring all UI elements loaded and functioned correctly
- Input validation – check how the solution responds to various types of input – valid, invalid, and edge cases (e.g., special characters, SQL injection)
- Functional requirements – ensure that login, sign-up, and booking behave as expected
- Error handling – check that the solution handles errors gracefully and returns appropriate output if an error occurs

Technical users were not permitted to access the solution's source code, only the same information that all users receive.

For this part, testers decided to use the approach of exploratory testing. Exploratory testing is an approach where testers simultaneously learn about the software, design tests, and execute them without predefined test cases, instead focusing on the tester's creativity for discovering defects in the solution.

Findings

Issue	Expected behaviour	Actual behaviour	Impact	Next steps
If users were to block client-side JavaScript execution, this would result in the solution no longer verifying inputs	The solution would still validate inputs on the back-end (server-side)	The solution no longer verifies any user inputs, including server-side	**Severe** – the solution freely accepts input regardless of validity	Implement server-side input validation and cleansing, especially where users can interact with the database
With blocked JavaScript, users could create accounts with invalid email addresses	A user inputting "test" as their email address should be flagged by the server and the user should be met with an error message	The solution accepts the email address as valid, and adds it to the database 	**Severe** – the solution freely accepts the input, this could be used maliciously, especially if the server-side code is susceptible to SQL injection	Implement server-side validation for email input
With blocked JavaScript, users were able to leave all text fields blanks	A user inputting blank form details should be flagged by the server and the user should be met with an error message	The solution accepts the blank form inputs as valid and adds it to the database	**Severe** – blank data isn't necessarily harmful, but the implications that this could input harmful data makes this a	Implement server-side validation for all user input

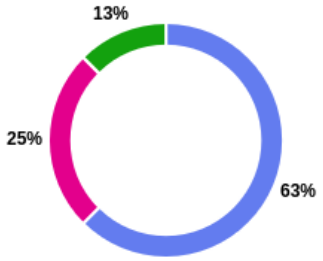


Graphs

2. What is your age?

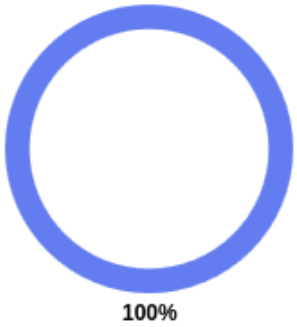
More details

Under 18	5
18-24	2
25-34	0
35-44	0
45-54	1



3. Were you able to navigate around the prototype?

Yes	10
No	0



4. Did the prototype allow you to create an account?

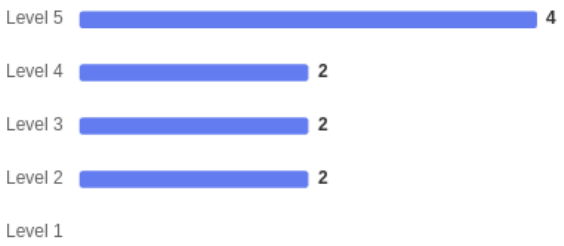
Yes	9
No	1

5. How clear were the password requirements on the Register form?

More details

3.80

Average Rating

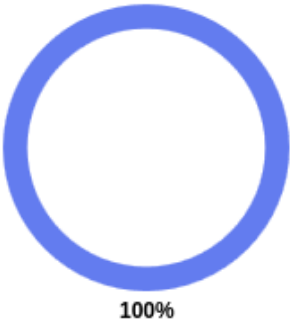


6. Did the prototype allow you to sign in?

Yes	9
No	1

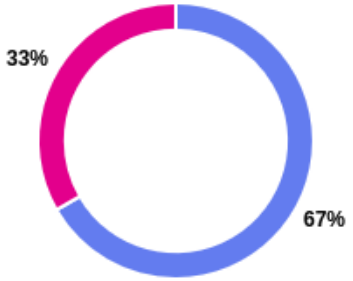
7. Were you able to access the account dashboard?

Yes	9
No	0



8. Was the information on the account dashboard displayed clearly?

Yes	6
No	3



9. Were you able to book a hotel stay?

Yes	9
No	1

10. How easy was it to book a hotel stay?

More details

8.11

Average Rating

11. (Hotel stay) Did you get a confirmation email?

Yes	1
No	8

12. Did your hotel booking(s) appear on the account dashboard?

Yes	8
No	1

13. Were you able to book zoo tickets?

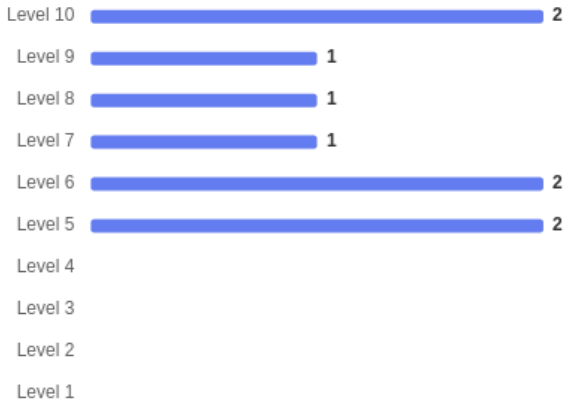
Yes	9
No	1

14. How easy was it to book zoo tickets?

More details

7.33

Average Rating



15. (Zoo tickets) Did you get a confirmation email?

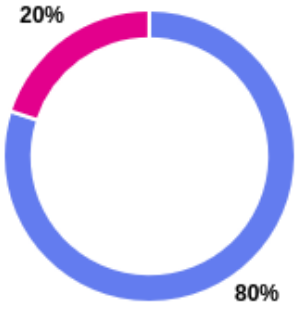
Yes	1
No	8

16. Did your zoo ticket(s) appear on the account dashboard?

Yes	8
No	1

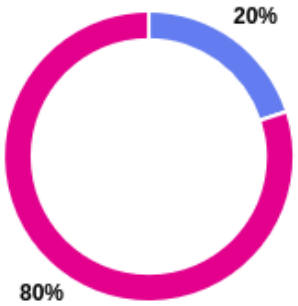
18. Did you use the prototype on a computer (desktop, laptop)?

Yes	8
No	2



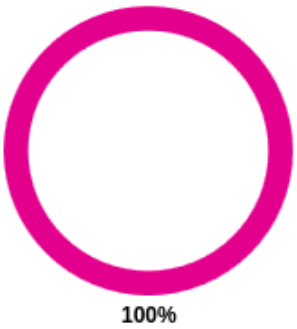
20. Did you use the prototype on a mobile device (phone, tablet)?

Yes	2
No	8



22. Were there any accessibility settings you could access?

Yes	0
No	10

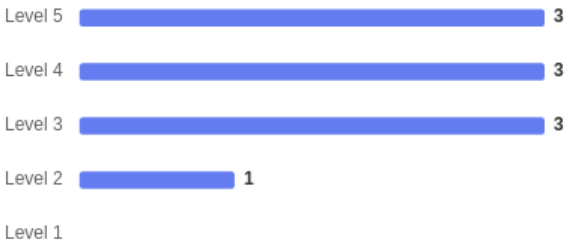


24. In your opinion, how appropriate was the colour theme of the prototype?

More details

3.80

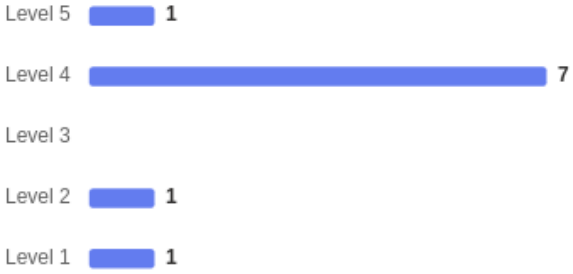
Average Rating



25. How intuitive are the layout and controls of the prototype?

3.60

Average Rating

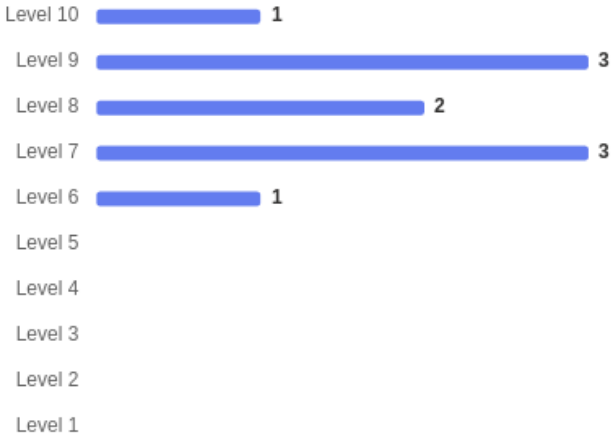


26. On a scale of 1 to 10, how would you rate the responsiveness of the prototype?

Mor

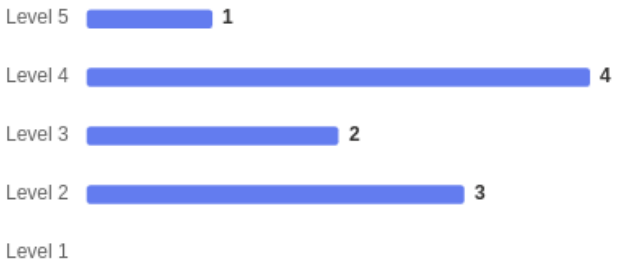
8.00

Average Rating



27. Overall, how would you rate your experience using the Riget Zoo Adventures website prototype?

Mo

3.30

Average Rating